



Global Food & Nutrition Database 2026

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Abstract

With the rapid growth of healthcare analytics and food awareness systems, data-driven solutions have become essential for understanding nutrition and improving healthy lifestyle decisions. This project presents a complete data analytics solution based on the *Global Food & Nutrition Database 2026* dataset collected from USDA FoodData Central and Open Food Facts.

The project focuses on transforming raw and unstructured nutritional datasets into interactive and meaningful dashboards using Python and Power BI. A Medallion Architecture approach was applied to organize the data pipeline into Bronze, Silver, and Gold layers, ensuring better data quality, consistency, and scalability.

The raw data was cleaned, transformed, and analyzed using Python libraries such as Pandas and NumPy. Missing values, duplicates, and inconsistent formats were handled before building analytical datasets and KPIs. The cleaned data was then visualized using Power BI dashboards that provide insights into nutrition values, health indicators, food allergens, and food category analysis.

The final system helps users explore food products, compare nutritional values, analyze allergen risks, and identify healthier food options through interactive visualizations and business intelligence techniques.

Table of Contents

Chapter 1: Project Overview

- 1.1 Introduction
- 1.2 Motivation
- 1.3 Project Objectives
- 1.4 Dataset Description

Chapter 2: Framework and Analysis

- 2.1 Tools and Technologies
- 2.2 Methodology (Medallion Architecture)
- 2.3 Analysis Requirements and KPIs

Chapter 3: Data Engineering and Implementation

- 3.1 Data Cleaning Process
- 3.2 Python Implementation
- 3.3 Data Transformation Results

Chapter 4: Design and Visualization

- 4.1 Dashboard Design and Layout
- 4.2 Power BI Dashboard Implementation
- 4.3 Dashboard Insights and Analysis

Chapter 5: Machine Learning Model Development and Evaluation

- 5.1 Machine Learning Overview
- 5.2 Dataset and Nutritional Features
- 5.3 Feature Selection
- 5.4 Training and Testing Process
- 5.5 Random Forest Regression Model
- 5.6 How the Prediction Process Works
- 5.7 Model Evaluation
- 5.8 Prediction Visualization
- 5.9 Actual and Predicted Results Comparison
- 5.10 Conclusion

Chapter 1: Project Overview

1.1 Introduction

In recent years, nutrition and healthcare analytics have become increasingly important due to the growing awareness of healthy lifestyles and food-related diseases. Large nutritional datasets provide valuable information, but raw data alone is difficult to understand and analyze without proper processing and visualization.

This project aims to build an intelligent food and nutrition analytics system using the *Global Food & Nutrition Database 2026*. The project combines data engineering, data analysis, and business intelligence techniques to transform raw nutritional data into interactive dashboards and meaningful insights.

Using Python for preprocessing and Power BI for visualization, the project provides an easy way to analyze food categories, nutritional values, allergen information, and health indicators.

1.2 Motivation

The motivation behind this project comes from several important challenges:

- Nutritional datasets are often large, messy, and difficult for normal users to understand.
- Many people need easier access to health and food information to support healthier food choices.
- Food allergies and unhealthy eating habits are increasing globally.
- Organizations need analytical systems that can transform raw food data into visual and understandable insights.

This project helps bridge the gap between raw nutritional data and decision-making through interactive dashboards and analytical KPIs.

1.3 Project Objectives

The main objectives of this project are:

- Clean and preprocess large nutritional datasets using Python.
- Build a structured data pipeline using Medallion Architecture.
- Create calculated health and nutrition KPIs.
- Analyze food categories and nutritional composition.
- Identify allergen distribution and safe food products.
- Design interactive dashboards using Power BI.

- Transform raw data into understandable business insights.

1.4 Dataset Description

The project uses the *Global Food & Nutrition Database 2026* dataset from Kaggle, which combines data from USDA FoodData Central and Open Food Facts.

The dataset includes:

- Food product names
- Food categories
- Calories and macronutrients
- Protein, fat, carbohydrates, sugar, and fiber values
- Health scores
- Brand names
- Allergen information
- Ingredient details

The dataset contains thousands of food products and nutritional records that support advanced analysis and visualization.

Chapter 2: Framework and Analysis

2.1 Tools and Technologies

Tools and technologies used to support data collection, preprocessing, analysis, visualization, documentation, and project. The tools are listed as following.

2.1.1 Python

Python was the main programming language used in this project due to its simplicity, flexibility, and strong support for data analysis. It was used for data preprocessing, cleaning, transformation, and exploratory data analysis (EDA).

Python also enabled efficient handling of large datasets and integration with visualization tools such as Power BI.

2.1.2 Pandas

Pandas is a Python library specialized in data manipulation and analysis. It was used extensively for:

- Reading CSV datasets
- Cleaning and filtering data
- Handling missing values
- Data transformation and preprocessing
- Organizing structured tables

Pandas provided efficient functions for managing large amounts of nutritional and food-related data.

2.1.3 NumPy

NumPy is a numerical computing library in Python used for mathematical and statistical operations. It supported:

- Numerical calculations
- Array operations
- Data standardization
- Statistical analysis

NumPy improved processing speed and simplified numerical computations during analysis.

2.1.4 Power BI

Power BI was used as the main visualization and dashboard development tool. It enabled the creation of interactive dashboards and visual reports to present insights from the dataset.

The dashboard included:

- Nutritional analysis visualizations
- Food comparison charts
- Health and EcoScore insights
- Interactive filters and KPIs

Power BI improved data interpretation and user interaction with the project results.

2.1.5 Kaggle Dataset

The dataset used in this project was obtained from Kaggle, specifically the Open Food Facts dataset. It contains detailed information about food products, including:

- Product names and categories
- Nutritional values
- Ingredients and allergens
- EcoScore and health-related data

The dataset provided a large real-world data source suitable for analysis and visualization.

2.1.6 CSV Files

The project data was stored and processed in CSV (Comma-Separated Values) format. CSV files are lightweight and widely used for structured datasets.

CSV files were used because they:

- Are easy to read and process
- Support integration with Python and Power BI
- Allow efficient storage of tabular data

2.1.7 Medallion Architecture

The project followed the Medallion Architecture approach for organizing data processing stages. This architecture improves data quality and workflow organization through multiple layers:

- Bronze Layer: Raw collected data
- Silver Layer: Cleaned and processed data
- Gold Layer: Final analytical and visualization-ready data

This structure improved data consistency and simplified dashboard development.

2.1.8 GitHub

GitHub was used for storing project files and managing source code. It allowed team members to collaborate efficiently and maintain version control during development.

GitHub also supported:

- File sharing
- Project backup
- Code organization
- Collaboration between team members

2.2 Methodology (Medallion Architecture)

To organize and process the data efficiently, the project follows the Medallion Architecture approach, which divides the workflow into three layers:

Bronze Layer (Raw Data)

This layer contains the original CSV files imported directly from the dataset source without modification. The data includes duplicates, missing values, and inconsistent formats.

Bronze Layer

Silver Layer (Cleaned Data)

In this layer, Python was used to:

- Remove duplicates
- Handle missing values

- Standardize data types
- Clean inconsistent columns
- Merge related tables
- Normalize nutritional units

This stage improved data quality and prepared the data for analysis.

Silver Layer

Gold Layer (Analytical Data)

The final layer contains transformed and analytical datasets optimized for Power BI dashboards. Additional KPIs and calculated measures were created in this layer to support reporting and visualization.

Gold Layer

2.3 Analysis Requirements and KPIs

Several analytical KPIs were developed to measure nutrition and health performance, including:

- Average calories
- Average protein
- Average fat
- Protein efficiency ratio
- Fiber-to-carbohydrate ratio
- High-calorie food ratio
- Health score indicators
- Safe products percentage
- Gluten, dairy, and nuts ratios

These KPIs help users evaluate food quality and compare different food categories.

Chapter 3: Data cleaning and Implementation

3.1 Data Cleaning Process

The raw dataset required several preprocessing steps before analysis. Using Python, the following operations were performed:

- Removing duplicated records
- Handling null and missing values
- Renaming inconsistent columns
- Filtering unnecessary fields
- Standardizing nutritional measurements
- Merging related datasets
- Creating calculated columns and health indicators

The cleaning process improved data consistency and reliability.

3.2 Python Implementation

Python was used as the main data engineering tool in this project. Important libraries included:

- Pandas
- NumPy
- Matplotlib
- Seaborn

The implementation focused on:

- Data preprocessing
- Feature engineering
- KPI generation
- Data transformation
- Exporting cleaned datasets for Power BI

Python scripts automated the transformation process and reduced manual effort significantly.

3.3 Data Transformation Results

After preprocessing, the dataset became more organized and analysis-ready. The final transformed dataset included:

- Clean nutritional information

- Health score calculations
- Food category classification
- Allergen flags
- Nutrition KPIs

The transformed data was then connected to Power BI for visualization and dashboard creation.

Chapter 4: Design and Visualization

4.1 Dashboard Design and Layout (Stitch)

The dashboard layout was designed using stitch to provide a simple and user-friendly experience. KPIs were placed at the top for quick insights, while charts and filters were organized clearly to improve usability and interaction.

The dashboards include:

-
- The landing page
- Nutrition Dashboard
- Health Dashboard
- Allergens Dashboard
- Cross Analysis Dashboard

Interactive slicers were added to allow users to filter results by food type, category, brand, allergens, and health score.

[design page](#)

4.2 Power BI Dashboard Implementation

Power BI was used to create interactive dashboards and visual analytics reports. Different visualizations were implemented, including:

- KPI Cards
- Bar Charts
- Pie Charts
- Scatter Plots
- Distribution Charts
- Comparative Analysis Visuals

DAX measures were also used to calculate custom metrics and improve dashboard performance.

4.3 Dashboard Insights and Analysis

Nutrition Dashboard

This dashboard analyzes nutritional composition and food categories. It highlights:

- Average calories, protein, and fat values
- High-calorie foods
- Protein efficiency measurements
- Comparison between food categories
- Relationships between calories and fat content

Health Dashboard

This section focuses on food health indicators and sugar analysis. It provides:

- Average health scores
- Healthy and low-health product comparison
- Sugar distribution by food category
- Calories analysis by food type

Allergens Dashboard

The allergens dashboard analyzes food safety and allergen distribution, including:

- Gluten ratio
- Dairy ratio
- Nuts ratio
- Safe products percentage
- Allergen distribution across products

Cross Analysis Dashboard

This dashboard combines nutrition and health data to identify relationships between food quality, calorie levels, and health indicators.

Processed Products Nutritional Analysis Dashboard

The processed products dashboard analyzes nutritional quality and food processing levels, including:

- Ultra-processed products ratio (NOVA Group 4)
- Nutri-Score distribution (A–E)
- High-quality products percentage (Nutri-Score A)
- Low nutrition products ratio (Nutri-Score D & E)
- Average calories and protein content
- Allergen presence across processing levels

- Relationship between calories and protein values
- Identification of high-calorie, low-nutrition products

Chapter 5: Machine Learning Model Development and Evaluation

5.1 Machine Learning Overview

Machine learning is a branch of artificial intelligence that enables computer systems to learn patterns from data and make predictions without being explicitly programmed for every task.

In this project, machine learning is used to predict the health score of food products based on their nutritional values. The model analyzes nutritional information such as calories, protein, fat, sugar, sodium, fiber, and vitamins to estimate how healthy a food item is.

The implemented machine learning system works by:

1. Receiving nutritional data as input
2. Learning relationships between nutrients and health scores
3. Training on historical food data
4. Generating predictions for new food items
5. Comparing predicted results with actual values

The model can:

- Predict food health scores automatically
- Analyze nutritional quality
- Support healthy food recommendation systems
- Detect unhealthy food patterns
- Assist in nutrition analysis applications

5.2 Dataset and Nutritional Features

The dataset used for training the machine learning model was imported from a cleaned CSV file named ``usda_clean_data.csv``. The dataset contains nutritional information for various food items and their corresponding health scores.

The following libraries were imported at the beginning of the implementation:

- NumPy for numerical computations
- Pandas for data manipulation
- Matplotlib and Seaborn for visualization
- Scikit-learn modules for preprocessing, model training, and evaluation

The dataset was loaded using the Pandas ``read_csv()`` function:

After loading the dataset, the first few rows were displayed using `df.head()` in order to inspect the structure and verify the successful loading of the data.

The dimensions of the dataset were checked using:

This step helped determine the number of rows and columns available for training and testing.

5.3 Feature Selection

The model uses multiple nutritional attributes as input features to predict the target variable, which is the health score.

The selected input features include:

- Calories
- Carbohydrates
- Calcium
- Fat
- Protein
- Saturated Fat
- Vitamin C
- Fiber
- Iron
- Sodium
- Sugar
- Cholesterol

The target variable is:

* Health Score

The feature and target variables were defined as follows:

Feature selection is an important step because it determines which nutritional factors contribute to predicting the overall health score.

5.4 Training and Testing Process

To evaluate the performance of the machine learning model effectively, the dataset was divided into training and testing subsets.

The `train_test_split()` function from Scikit-learn was used with the following configuration:

- * 80% of the data for training
- * 20% of the data for testing
- * Random state set to 42 for reproducibility

Description:

This approach ensures that the model is evaluated on unseen data, which helps measure its generalization capability.

Data Standardization

Before training the machine learning model, feature scaling was applied using the `StandardScaler` class.

Standardization transforms the data so that each feature has:

- Mean equal to 0
- Standard deviation equal to 1

This preprocessing step improves numerical stability and ensures that all features contribute equally during training.

Description:

The scaler was fitted only on the training data and then applied to both training and testing datasets to prevent data leakage.

5.5 Random Forest Regression Model

The selected machine learning algorithm for this project is the Random Forest Regressor.

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

The model was initialized with:

100 decision trees (`n_estimators=100`)

Description:

The Random Forest algorithm was selected because it:

- Handles complex relationships between features
- Provides strong prediction performance
- Is resistant to overfitting
- Works well with numerical datasets

5.6 How the Prediction Process Works

After preprocessing the data, the model was trained using the scaled training dataset.

Description:

During training, the Random Forest algorithm learns patterns and relationships between nutritional values and health scores.

The training process involves:

1. Creating multiple decision trees
2. Training each tree on random subsets of the data
3. Combining predictions from all trees
4. Producing the final averaged prediction

This ensemble strategy improves model robustness and prediction accuracy.

Prediction and Testing

Once training was completed, predictions were generated using the testing dataset.

Description:

The predicted values represent the estimated health scores generated by the trained machine learning model.

These predictions were then compared with the actual health scores from the dataset.

5.7 Model Evaluation

The performance of the model was evaluated using the R-squared (R^2) score.

The R^2 score measures how well the predicted values match the actual values. Its value ranges from 0 to 1:

- A value close to 1 indicates strong predictive performance
- A value close to 0 indicates poor predictive capability

Description:

The evaluation results demonstrated the effectiveness of the Random Forest Regressor in predicting food health scores using nutritional information.

5.8 Prediction Visualization

To visually analyze the model performance, a scatter plot was created to compare actual and predicted values.

The visualization included:

- Scatter points representing predictions
- A reference line representing perfect predictions

Description:

The closer the scatter points are to the diagonal reference line, the more accurate the model predictions are.

5.9 Actual and Predicted Results Comparison

A comparison table was generated to display the actual health scores alongside the predicted values.

Description:

This comparison provides a direct view of prediction accuracy and helps identify prediction differences or outliers.

The comparison table is useful for:

- Verifying prediction quality

- Detecting large prediction errors
- Understanding model behavior on different samples

ML model

5.10 Conclusion

This project successfully demonstrated how data engineering and business intelligence techniques can transform raw nutritional datasets into meaningful and interactive analytical dashboards.

By using Python for preprocessing and Power BI for visualization, the project provided valuable insights into food nutrition, health indicators, and allergen analysis. The Medallion Architecture approach improved data organization and quality, while the dashboards simplified the exploration of complex nutritional information.